

Scheduling Quantum Annealing for Active User Detection in a Non-Orthogonal Multiple Access Network

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Outline

- ① Introduction
- ② QA formulation for active user detection
- ③ QA scheduling
- ④ Mean gap
- ⑤ Conclusion



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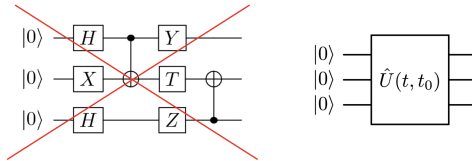
Statistical inference

- In wireless communications, several signal processing tasks belong to the field of *statistical inference*
An inference problem ¹ consists in deducing the values of some variables $\{x_i\}_{i=1,\dots,N}$ from observed data $\{y_j\}_{j=1,\dots,M}$
- Each attempt of reconstruction can be translated into a minimization problem $\rightarrow$ Bayesian inference)
Maximum likelihood, maximum a posteriori
- When $\{x_i\}_{i=1,\dots,N}$ are binary variables, one faces multivariate binary optimization which is cumbersome to perform

¹Lenka Zdeborová and Florent Krzakala, "Statistical physics of inference: Thresholds and algorithms"

Principle of quantum annealing

- Quantum Annealing (QA) ² is a promising quantum analog algorithm for ground state searching of Ising models



- GS searching of Ising models $\equiv$ *quadratic unconstrained binary optimization* (QUBO) problems $\rightarrow$ opens wide range of perspectives for industrial applications
- A QA approach must be adapted for each set of observed data $\{y_j\}_{j=1,\dots,M}$... Is it convenient ?

²T. Kadowaki and H. Nishimori, "Quantum annealing in the transverse Ising model"



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Context: active user detection

In massive wireless networks, resource allocation prior to transmission is no longer adapted for 5G applications



Transmission protocol ³:

- ① (Downlink) The AP indicates the beginning of the frame
- ② (Uplink) Active users send an identification code
- ③ AP estimates the active users to start an allocation protocol

³Jeannerot et al. Mitigating User Identification Errors in Resource Optimization for Grant-Free Random Access

Signal model

A random subset of active users transmit their code c_i to the access point

$$\mathcal{N}_A = \left\{ i \in \{1, \dots, N\} \text{ s.t. } b_i^{(0)} = 1 \right\} \quad (1)$$

Received signal:

$$\mathbf{y} = \tilde{\mathbf{C}} \cdot \mathbf{b}^{(0)} + \mathbf{n} \quad (2)$$

where:

$$\begin{aligned} \tilde{\mathbf{C}} &= [w_1 \mathbf{c}_1 \cdots w_N \mathbf{c}_N] \in \mathbb{C}^{M \times N} \text{ with } M < N \\ w_i &\sim \mathcal{N}(0, 1), \quad i = 1, \dots, N \text{ (Channel coefficients)} \\ n_j &\sim \mathcal{N}(0, \xi^2) \quad j = 1, \dots, M \text{ (Additive noise)} \end{aligned} \quad (3)$$

The identification codes are defined in preamble



Maximum a posteriori detection

Received signal:

$$\mathbf{y} = \tilde{\mathbf{C}} \cdot \mathbf{b}^{(0)} + \mathbf{n} \quad (2)$$

The codes are non-orthogonal. The most reliable way to recover the *activity pattern* $\mathbf{b}^{(0)}$ is to compute the MAP detector:

$$\hat{\mathbf{b}}^{\text{MAP}} = \arg \max_{\mathbf{b} \in \{0,1\}^N} \mathbb{P}(\mathbf{b} | \mathbf{y}, \mathbf{w}) \quad (4)$$

Assuming the prior $\mathbf{b} \sim \mathcal{U}([0,1]^N)$, one has:

$$\hat{\mathbf{b}}^{\text{MAP}} = \arg \min_{\mathbf{b} \in \{0,1\}^N} \left\{ \left\| \mathbf{y} - \tilde{\mathbf{C}} \cdot \mathbf{b} \right\|_2^2 \right\} \quad (5)$$

Ising Hamiltonian

The previous optimization problem is exactly a QUBO problem! It can be recasted as:

$$\hat{\mathbf{b}}^{\text{MAP}} = \arg \min_{\mathbf{b} \in \{0,1\}^N} \left\{ \sum_{i,j} Q_{ij} b_i b_j + \sum_i a_i b_i \right\}. \quad (6)$$

In terms of spin variables $\sigma \in \{\uparrow, \downarrow\}^N$, we obtain the following problem Hamiltonian:

$$H_P = - \sum_{i < j} J_{ij} \sigma_i \sigma_j - \sum_i h_i \sigma_i, \quad (7)$$

with:

$$\begin{cases} h_i &= -w_i (\mathbf{c}_i \cdot \tilde{\mathbf{y}}) \\ J_{ij} &= -\frac{1}{2} (w_i w_j) (\mathbf{c}_i \cdot \mathbf{c}_j) \end{cases}. \quad (8)$$



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Reminder: general idea of QA

We consider the global Hamiltonian:

$$\hat{H}(u(t)) = (1 - u(t))\hat{H}_P + u(t)\hat{H}_C \quad (9)$$

The term u is called the *control function* and obeys:

$$u(0) = 1, \quad u(T) = 0 \quad (10)$$

- $\hat{H}_P$ is the *problem Hamiltonian*. It contains the objective function ($\hat{H}_P \equiv \log\text{-likelihood}$)
- $\hat{H}_C$ is a transverse term called the *control Hamiltonian*. It introduces fluctuations between states. In this work:

$$\hat{H}_C = - \sum_i \hat{\sigma}_i^x \quad (11)$$

Adiabatic approximation (1/2)

We consider the global Hamiltonian:

$$\hat{H}(u(t)) = (1 - u(t))\hat{H}_P + u(t)\hat{H}_C \quad (9)$$

Let us denote $\{\varepsilon_n(t), |\phi_n(t)\rangle\}$ the instantaneous eigensystem of $\hat{H}(t)$

Definition

The spectral gap of the system is:

$$\Delta(t) = \varepsilon_1(t) - \varepsilon_0(t) \quad (12)$$

It depends on t only through $u(t)$.

Adiabatic approximation (2/2)

We consider the global Hamiltonian:

$$\hat{H}(u(t)) = (1 - u(t))\hat{H}_P + u(t)\hat{H}_C \quad (9)$$

Assume that the system is initialized to $|\Psi(0)\rangle = |+\rangle^{\otimes N}$.

Adiabatic approximation

An adiabatic approximation ensures the stability of the ground state of a system subject to perturbations. In this work, we use:

$$\left| \frac{du}{dt} \right| \frac{|\langle \hat{H}_C - \hat{H}_P \rangle_{1,0}(u)|}{\Delta(u)^2} \ll 1 \quad \forall t \in [0, T] \Rightarrow P_{\text{QA}}(T) \approx 1, \quad (13)$$

where we denoted:

$$P_{\text{QA}}(t) = |\langle \phi_0(T) | \psi(t) \rangle|^2. \quad (14)$$

QA scheduling: choice of the control function (1/2)

$$\left| \frac{du}{dt} \right| \frac{|\langle \hat{H}_C - \hat{H}_P \rangle_{1,0}(u)|}{\Delta(u)^2} \ll 1 \quad \forall t \in [0, T] \Rightarrow P_{QA}(T) \approx 1 \quad (13)$$

- **Linear control function:**

$$u(t) = 1 - \frac{t}{T_{\text{lin}}} \Rightarrow T_{\text{lin}} \ll \Delta_{\text{min}}^{-2} \quad (15)$$

But due to first order quantum phase transition, $T_{\text{lin}} \sim e^{-\lambda N}$

- **Optimized control function:** take $\epsilon > 0$ and numerically solve:

$$\frac{du}{dt} = -\epsilon \Delta^2(u) \Rightarrow T = \frac{1}{\epsilon} \int_0^1 \frac{du}{\Delta^2(u)} \quad (16)$$

In some examples, it allows to reduce the constant λ .

QA scheduling: choice of the control function (2/2)

- **Optimized control function:** take $\epsilon > 0$ and numerically solve:

$$\frac{du}{dt} = -\epsilon \Delta^2(u) \implies T = \frac{1}{\epsilon} \int_0^1 \frac{du}{\Delta^2(u)} \quad (16)$$

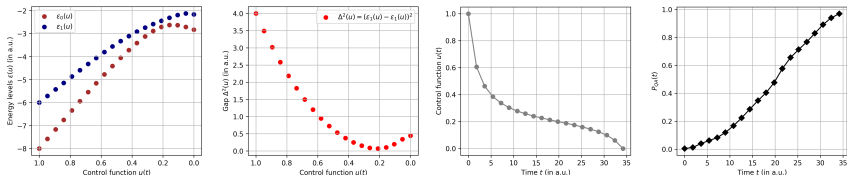


Figure 1: Spectral gap, appropriated control function and resulting $P_{QA}(t)$ ($\epsilon = 0.1$)



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Generic gap

In our context, a QA protocol depends on the observed data $(\mathbf{y}, \mathbf{w})$.

$J_{(\mathbf{y}, \mathbf{w}), ij}, h_{(\mathbf{y}, \mathbf{w}), i}$: easy to get

$\Delta_{(\mathbf{y}, \mathbf{w})}(u)$: hard to get

Once the problem Hamiltonian is defined, we propose to use the *mean gap*.

Definition: mean gap

The mean gap is computed by an average over the channel coefficients $\mathbf{w}$ and the initial activity patterns $\mathbf{b}^{(0)}$, assuming $\mathbf{n} = 0$:

$$\Delta_{(\text{mean}), N}^2(u) = \mathbb{E}_{(\mathbf{y}, \mathbf{w})} \left(\Delta_{(\mathbf{y}, \mathbf{w})}^2 \right) = \sum_{\mathbf{b}^{(0)} \in \{0,1\}^N} \int d\mathbf{w} f(\mathbf{b}^{(0)}, \mathbf{w}) \Delta_{(\mathbf{y}, \mathbf{w})}^2(u). \quad (17)$$

Generic gap

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This mean gap allows to compute a generic control function for $\epsilon > 0$:

$$\begin{cases} \frac{du_{(\text{mean}),N}}{dt} &= -\epsilon \Delta_{(\text{mean}),N}^2(u_{(\text{mean}),N}(t)) \\ u_{(\text{mean}),N}(0) &= 1 \end{cases} \quad (18)$$

Generic gap

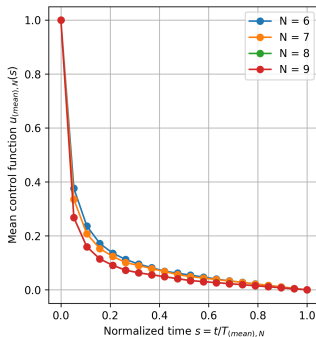


Figure 2: Shape of the mean control functions with fading coefficients in the network obtained for $N = 6, \dots, 9$ with respect to $s = t/T_{(\text{mean}),N}$

Reliability evaluation

We define the following random variables:

$$P_{QA} (T_{(\text{mean}),N}; (\mathbf{y}, \mathbf{w})) \equiv P_{QA}^{(\text{mean})}(\mathbf{y}, \mathbf{w})$$

$$P_{QA} (T_{(\mathbf{y}, \mathbf{w})}; (\mathbf{y}, \mathbf{w})) \equiv P_{QA}^{(\text{opti})}(\mathbf{y}, \mathbf{w}) \quad (19)$$

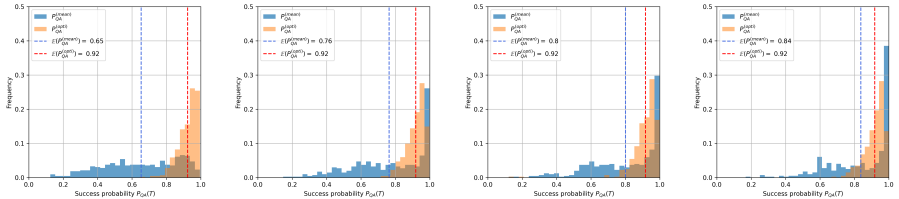


Figure 3: $P_{QA}^{(\text{mean})}$ (in blue) and $P_{QA}^{(\text{opti})}$ (in red) for $N = 8$ and $\text{SNR} = 15\text{dB}$

$$\epsilon^{(\text{opti})} = 0.100, \epsilon^{(\text{mean})} \in \{0.100, 0.050, 0.033, 0.025\},$$

Advantage over linear control function

We define the following random variables:

$$P_{QA} (T_{(\text{mean}),N}; (\mathbf{y}, \mathbf{w})) \equiv P_{QA}^{(\text{mean})}(\mathbf{y}, \mathbf{w})$$

$$P_{QA} (T_{(\text{lin}),N} = T_{(\text{mean}),N}; (\mathbf{y}, \mathbf{w})) \equiv P_{QA}^{(\text{lin})}(\mathbf{y}, \mathbf{w}). \quad (20)$$

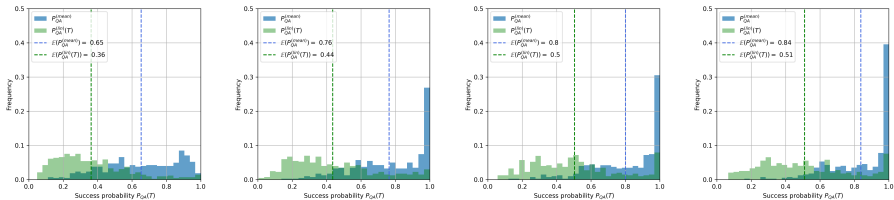


Figure 4: $P_{QA}^{(\text{mean})}$ (in blue) and $P_{QA}^{(\text{lin})}$ (in green) for $N=8$ and $\text{SNR}=15\text{dB}$

$$\epsilon^{(\text{mean})} \in \{0.100, 0.050, 0.033, 0.025\},$$



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Conclusion

Contributions

- Generic procedure to parameterize a QA process despite the randomness of observed data in the associated model
- Demonstrated reliability and advantage over a linear control function

Next steps

- Scalability of the mean annealing time with N (QMC methods)
- Deeper study of the sensitivity to the SNR



Thank you! Questions ?